



Homework 1 solution

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Problem 1

Let x_i be the number of shares sold of stock i , for $i = 1, \dots, 10$.

The bounds are

$$0 \leq x_i \leq 100.$$

For each sold share, the net cash kept today is

$$\text{net}_i = c_i - 0.30(c_i - p_i) - 0.01c_i,$$

where p_i is the purchase price and c_i is the current price.

The expected one-year value lost from selling x_i shares is $f_i x_i$, where f_i is the expected price in one year. The starting portfolio is fixed, so maximizing the expected value left in the account is the same as minimizing the expected value sold.

The linear program is

$$\begin{aligned} \min \quad & \sum_{i=1}^{10} f_i x_i \\ \text{subject to} \quad & \sum_{i=1}^{10} \text{net}_i x_i = 30000, \\ & 0 \leq x_i \leq 100, \quad i = 1, \dots, 10. \end{aligned}$$

The optimal sales are:

Stock	Shares sold	Shares remaining
1	0.000000	100.000000

Stock	Shares sold	Shares remaining
2	0.000000	100.000000
3	100.000000	0.000000
4	100.000000	0.000000
5	0.000000	100.000000
6	63.750749	36.249251
7	0.000000	100.000000
8	100.000000	0.000000
9	100.000000	0.000000
10	100.000000	0.000000

These sales raise exactly \$30,000.00 after taxes and transaction costs.
The expected before-tax one-year value of the remaining portfolio is \$20,893.71.

Problem 2

Yes, the problem can be transformed into a linear program.

Define

$$y = x_2^2.$$

Because $x_2 \geq 0$, every $y \geq 0$ corresponds to $x_2 = \sqrt{y}$. The constraint $x_2^2 - x_3 = 0$ becomes

$$y - x_3 = 0.$$

The transformed problem is

$$\begin{aligned}
& \min && 2x_1 + y + x_3 \\
& \text{subject to} && y - x_3 = 0, \\
& && 5x_1 + 3x_3 \leq 5, \\
& && x_1, y, x_3 \geq 0.
\end{aligned}$$

To write it in equality standard form, add a slack variable $s \geq 0$:

$$\begin{aligned}
& \min && 2x_1 + y + x_3 \\
& \text{subject to} && y - x_3 = 0, \\
& && 5x_1 + 3x_3 + s = 5, \\
& && x_1, y, x_3, s \geq 0.
\end{aligned}$$

Problem 3

Part (a)

Let x_{ij} equal 1 if worker i is assigned to project j , and 0 otherwise.
The cost for an assignment is \$20 times the number of hours.

The linear program is

$$\begin{aligned}
& \min && 20 \sum_{i=1}^4 \sum_{j=1}^4 h_{ij} x_{ij} \\
& \text{subject to} && \sum_{j=1}^4 x_{ij} = 1, \quad i = 1, \dots, 4, \\
& && \sum_{i=1}^4 x_{ij} = 1, \quad j = 1, \dots, 4, \\
& && 0 \leq x_{ij} \leq 1, \quad i, j = 1, \dots, 4.
\end{aligned}$$

Here h_{ij} is the number of hours required when worker i does project j .

Part (b)

The Python linear-programming solution already has 0-1 values:

Worker	Assigned project	Hours	Cost
1	2	3	\$60
2	1	5	\$100
3	3	7	\$140
4	4	8	\$160

The minimum total cost is

$$60 + 100 + 140 + 160 = \$460.$$

Problem 4

Let:

- b_t be units bought at the start of month t .
- q_t be units sold at the start of month t .
- I_t be inventory after the month- t buying or selling decision.
- $I_0 = 2000$ be the initial inventory.

I use the timing convention that the inventory cost i_t is charged on the inventory carried into month t , namely I_{t-1} . With that convention, the linear program is

$$\begin{aligned}
 \max \quad & \sum_{t=1}^{12} s_t q_t - \sum_{t=1}^{12} p_t b_t - \sum_{t=1}^{12} i_t I_{t-1} \\
 \text{subject to} \quad & I_t = I_{t-1} + b_t - q_t, \quad t = 1, \dots, 12, \\
 & I_0 = 2000, \\
 & I_{12} = 0, \\
 & 0 \leq I_t \leq 10000, \quad t = 0, \dots, 12, \\
 & b_t \geq 0, \quad q_t \geq 0, \quad t = 1, \dots, 12.
 \end{aligned}$$

If the course interprets the inventory cost as being charged after the month- t trade, replace I_{t-1} in the objective with I_t . The balance constraints remain the same.

Problem 5

Let:

- x_i be units purchased of bond i .
- y_1 be excess cash after paying the year-1 liability.
- y_2 be excess cash after paying the year-2 liability.

The reinvestment rate is 2%, so one dollar of surplus becomes 1.02 dollars one year later.

The linear program is

$$\begin{aligned} \min \quad & 102x_1 + 99x_2 + 98x_3 \\ \text{subject to} \quad & 105x_1 + 3.5x_2 + 3.5x_3 - y_1 = 12000, \\ & 103.5x_2 + 3.5x_3 + 1.02y_1 - y_2 = 18000, \\ & 103.5x_3 + 1.02y_2 = 20000, \\ & x_1, x_2, x_3, y_1, y_2 \geq 0. \end{aligned}$$

The Python solution is:

Variable	Value
x_1	102.265208
x_2	167.378469
x_3	193.236715
y_1	0.000000
y_2	0.000000

The minimum cost is \$45,938.72.

Allowing excess cash generation and reinvestment can help because surplus cash in an early period is not wasted. It earns interest and can pay later liabilities. For this data set, the optimal solution uses no surplus cash, so reinvestment does not lower the minimum cost.

Problem 6

Use path variables:

- m : flights from Calgary to Quebec City through Winnipeg and Montreal.
- h : flights from Calgary to Quebec City through Winnipeg and Hamilton.
- r : flights from Calgary to Quebec City through Winnipeg and Toronto.

The linear program is

$$\begin{array}{ll}\max & m + h + r \\ \text{subject to} & m + h + r \leq 5 \quad \text{Calgary to Winnipeg,} \\ & m \leq 4 \quad \text{Winnipeg to Montreal,} \\ & h \leq 5 \quad \text{Winnipeg to Hamilton,} \\ & r \leq 2 \quad \text{Winnipeg to Toronto,} \\ & m \leq 2 \quad \text{Montreal to Quebec City,} \\ & h \leq 1 \quad \text{Hamilton to Quebec City,} \\ & r \leq 3 \quad \text{Toronto to Quebec City,} \\ & m, h, r \geq 0.\end{array}$$

The optimal schedule is:

Path	Daily flights
Calgary-Winnipeg-Montreal-Quebec City	2
Calgary-Winnipeg-Hamilton-Quebec City	1
Calgary-Winnipeg-Toronto-Quebec City	2

The maximum number of daily flights from Calgary to Quebec City is 5.